As a student from another city, finding suitable hostel accommodation was never straightforward. That problem became the foundation of this project — a fully functional C++ console application that helps students find and book hostels while giving hostel owners a simple way to manage their records.

- **Dynamic Pricing Engine** — Built a real-time algorithm that calculates total fees based on room capacity, optional amenities such as Mess and Washroom, and luxury charges for prime location tiers.
- **Data Structuring** — Used custom structs and arrays to manage separate databases for different hostel blocks, keeping records organized without any external dependencies.
- **Input Validation** — Implemented strict type-checking to handle edge cases such as preventing runtime errors when non-numeric characters are entered.
- **Automated Billing** — Developed a logic engine that generates instant, detailed billing breakdowns — replacing manual calculation entirely.

Skills Applied: C++ · Data Structures · Conditional Logic · Input Validation · Software Architecture

Also uploaded on GitHub: GitHub - 2502272/HOSTEL-REQUIREMENT-SYSTEM-CPP: A C++ console application that automates hostel room allocation and fee calculation. Features include gender-based filtering, dynamic room configuration (1-4 seater), and automated receipt generation with real-time cost estimation. · GitHub